

Joystick Based Stepper Motor Angle Controller Using AVR MCU



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A stepper motor is associated with parameters like speed, direction, number of revolutions and angle of rotation. There are different types of stepper motor controllers designed to control required parameters as per specific applications. Here is a project for controlling a stepper motor's angle of rotation up to 360°.

Controlling the angle of rotations of stepper motors finds many applications including the following:

Satellite dish antenna positioning. A satellite dish antenna should be in alignment with the satellite in space to receive signal at maximum strength. Therefore it is important to position the dish antenna at required angle.

Aileron, elevator and rudder positioning in aircraft. In aircraft, it is required to position the aileron,

elevator and rudder at desired angles to increase or decrease lift, change direction and increase or decrease speed.

Rudder positioning on ship. To move the ship in a specific direction, it is important to set the angle of the rudder located in front of the propeller.

Tank guns or howitzers positioning. For hitting the bulls-eye (target), the tank gun (or howitzer) must be positioned at a specific angle.

One way to control the stepper motor's angle of rotation is by using a joystick. You can turn the joystick shaft at a specific angle to rotate the motor at the desired angle and, at the same time, setting of the angle of rotation can be displayed on the LCD screen.

As the joystick shaft is turned right or left, the motor immediately follows the joystick and rotates in the same direction by same angle.

That is, if the shaft is turned counter-clockwise, the motor also rotates in that direction through a similar angle, and vice versa.

Circuit and working

Circuit of the joystick based stepper motor angle controller using an AVR microcontroller (MCU) is shown in Fig. 1. It is built around AVR MCU ATmega16 (IC4), current driver chip ULN2003 (IC3), a 16x2 LCD module (LCD1) and a few other components.

In this project we use a 10-kilo-ohm joystick potmeter (JP1), which varies the voltage given as input to the built-in analogue-to-digital (ADC) converter of the MCU. The MCU, through the program, rotates the stepper motor at a desired angle per the position of the joystick shaft.

JP1 is connected as input to the first channel of the built-in ADC at pin PA0. It is connected in such a way that when its shaft is rotated fully, voltage at PA0 increases from 0V to 5V.

PORTB pins PB0 through PB3 are connected to input pins of IC3. These pins drive stepper motor M1. Output of IC3 drives four coils of the unipolar stepper motor. The common terminal (point E) of motor coils and common pin 9 of IC3 are connected to +5V.

PORTD of IC4 drives eight data pins

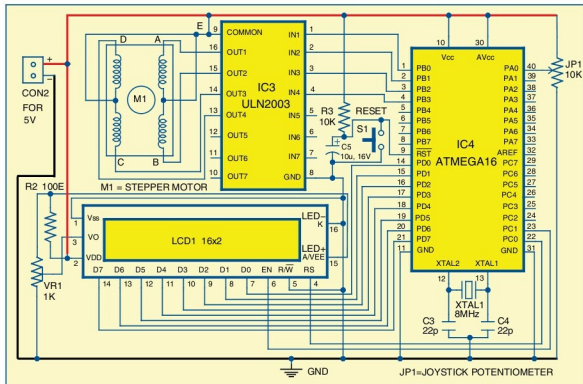


Fig. 1: Circuit diagram of the joystick based stepper motor angle controller using an AVR MCU